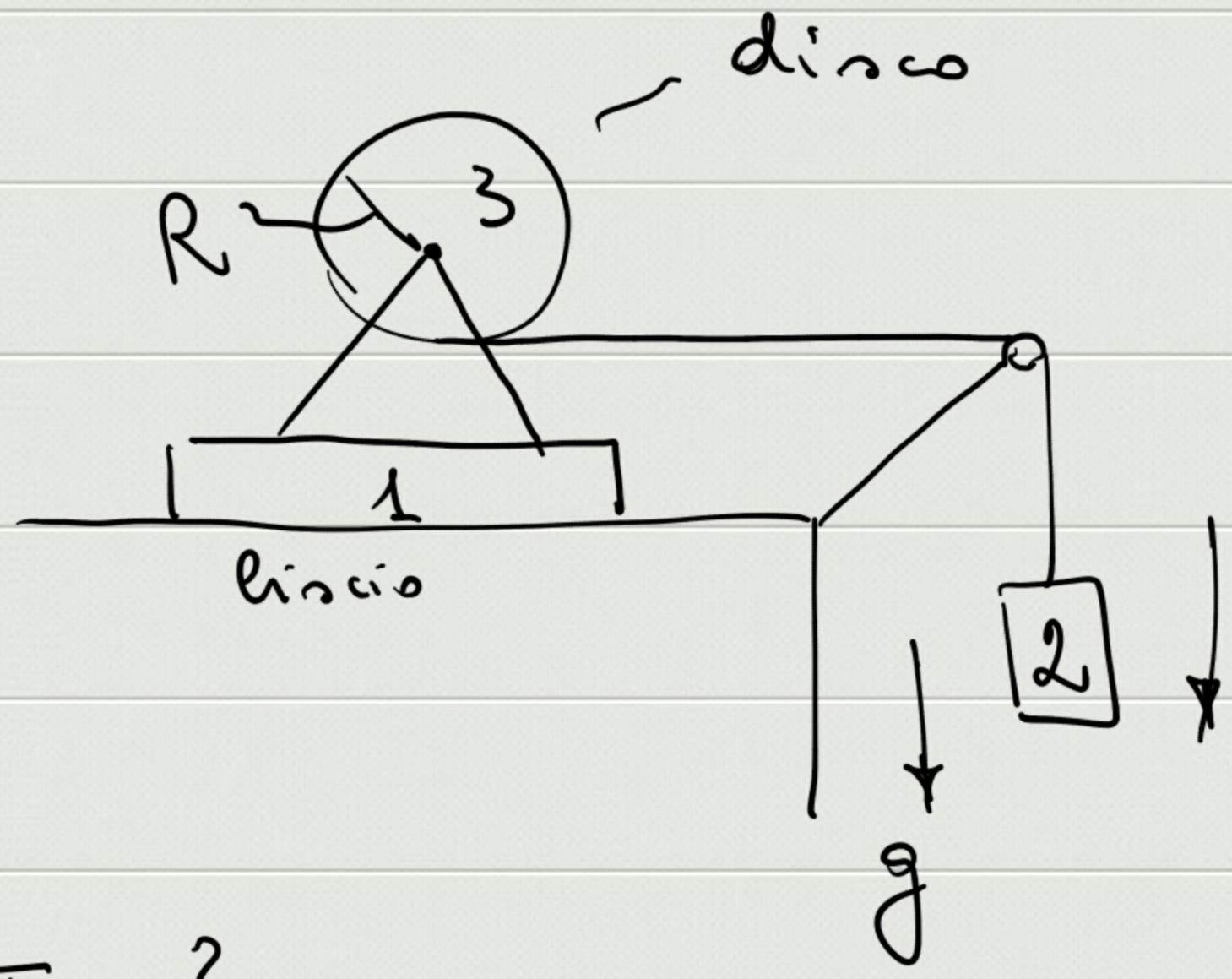


$m_1 = 2.5\,kg$

$m_2 = 1\,kg$

$m_3 = 2\,kg$

$R = 0.1\,m$

$T = ?$

$$\boxed{m_2 g - T = m_2 a_2}$$

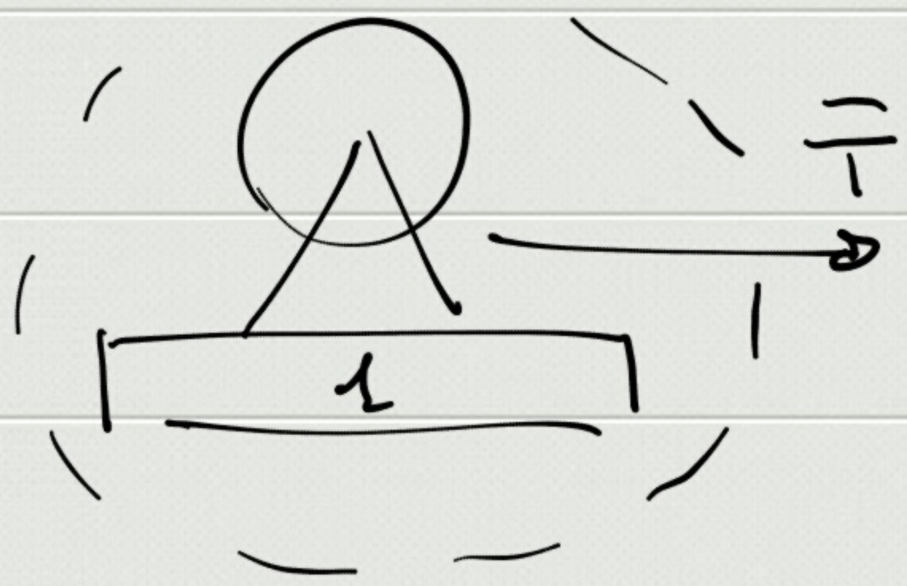
$$\boxed{T = (m_1 + m_3) a_{13}} \quad *$$

$$\boxed{F_{13} = ?}$$

$$\begin{cases} \bar{T} + \bar{F}_{13} + m_3 \bar{g} = m_3 \bar{a}_{13} \\ \bar{F}_{31} + m_1 \bar{g} + \bar{N}_1 = m_1 \bar{a}_{13} \end{cases} \quad *$$

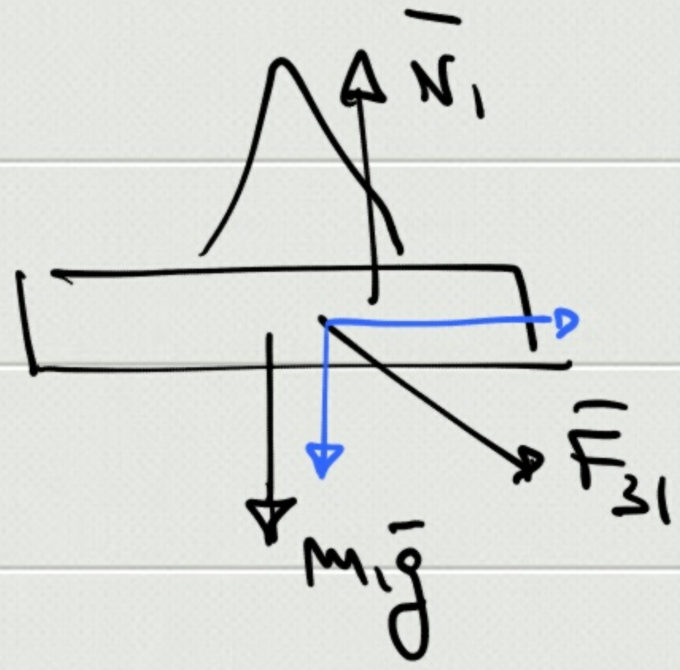
$$T = (m_1 + \cancel{m_3})(\cancel{a_{13}} + \alpha R) \qquad \alpha = \frac{a_{13}}{R}$$

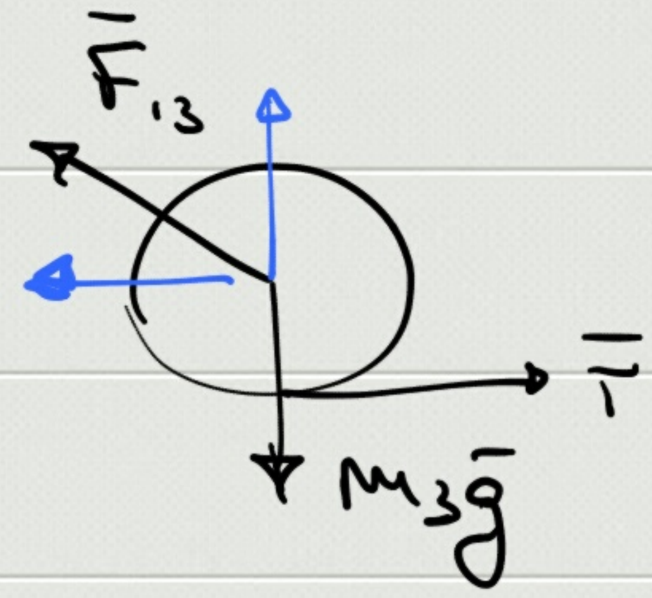
$$T = \cancel{m_3 a_{13}}$$



$$\boxed{\bar{R}^E = m_{TOT}\, \bar{a}_{CM}}$$

$$\Rightarrow T = (m_1 + m_3) a_{13}$$

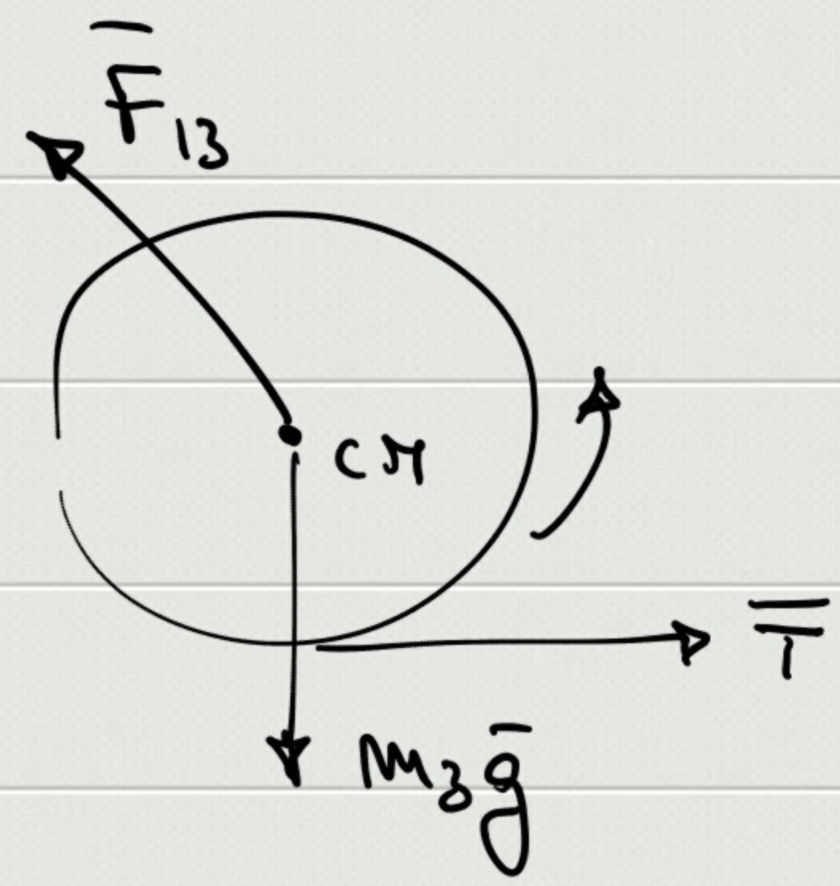




$$\cancel{RT + mgR = I\alpha}$$

$$\cancel{RT - mgR = I\alpha}$$

$$\cancel{R}$$



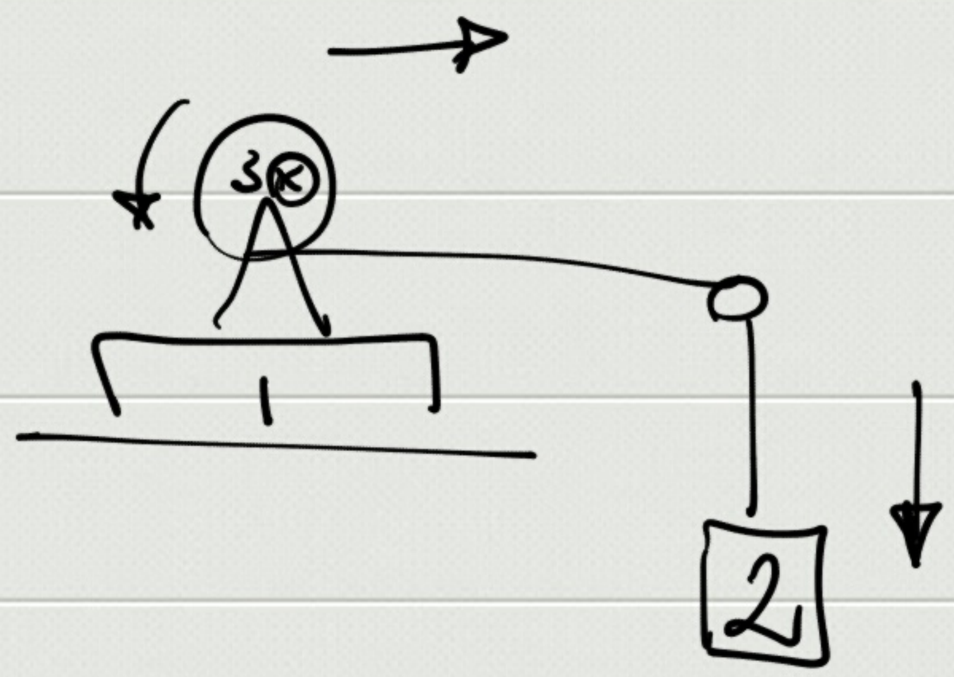
$$\bar{M}^{E}_{polo} = I\bar{\alpha} \Rightarrow \boxed{RT = I\alpha \quad \hookrightarrow \frac{1}{2} m_3 R^2}$$

$$a_2 = a_{13}$$

$$a_2 = \alpha R$$

$$\boxed{a_2 = a_{13} + \alpha R} \quad *$$

$$a_2 = a_{13} - \alpha R$$



$$a_{13} = 0 \Rightarrow \underbrace{a_2 = \alpha R}$$

$$\alpha = 0 \Rightarrow \underbrace{a_2 = a_{13}}$$

$$\begin{cases} m_2 g - T = m_2 a_2 \\ T = (m_1 + m_3) a_{13} \\ \cancel{R}T = \frac{1}{2} m_3 R^{\cancel{2}} \alpha \\ a_2 = a_{13} + \alpha R = \frac{T}{m_1 + m_3} + \frac{2T}{m_3} \end{cases}$$

$$\Rightarrow m_2 g = T + \left(\frac{m_2}{m_1 + m_3} + \frac{2 m_2}{m_3} \right) T$$

$$\Rightarrow T = \frac{m_2 g}{1 + \frac{m_2}{m_1 + m_3} + \frac{2 m_2}{m_3}} = 4.5 \text{ N}$$

$F_{13,x} = -2.5$ N

$F_{13,y} = 19.6$ N